

OG-RMM Platform: Gap Analysis, Improvement Recommendations & Competitive Comparison

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Executive Summary

The OG-RMM Platform has evolved through three major development cycles into a comprehensive, cloud-native remote monitoring and management system for oil and gas operations. This document provides a candid assessment of remaining capability gaps, a prioritized improvement roadmap, and a detailed comparison against the leading commercial platforms in the market.

The platform currently delivers **13 functional modules** across a polyglot microservices architecture (Go, Rust, Python, TypeScript), with full coverage of WT Petrotech's 14 product lines. The competitive analysis reveals that the platform matches or exceeds commercial alternatives in several dimensions — particularly real-time streaming architecture, ML-driven predictive maintenance, and financial ledger integration — while trailing in areas such as historian depth, certified safety system integration, and mobile field operations.

Part I: Remaining Gaps

1.1 Gap Inventory

Despite reaching 100% coverage of WT Petrotech's stated product lines, a rigorous technical review reveals **11 residual gaps** across five categories. These are not product-

line gaps but rather depth-of-implementation gaps — areas where the platform has surface coverage but lacks the production-grade depth expected by enterprise operators.

#	Gap	Category	Severity	Estimated Effort
G-01	No certified historian (PI System equivalent)	Data Infrastructure	High	8–12 weeks
G-02	No mobile field operations app	Field Operations	High	6–8 weeks
G-03	No IEC 62443 cybersecurity certification path	Security	High	12–16 weeks
G-04	No regulatory reporting automation (API 14C, 30 CFR 250)	Compliance	High	6–8 weeks
G-05	No digital twin / physics-based simulation	Advanced Analytics	Medium	10–14 weeks
G-06	No multi-tenant operator isolation	Architecture	Medium	4–6 weeks
G-07	No SIL-rated safety instrumented system (SIS) integration	Safety	Medium	8–10 weeks
G-08	No reservoir simulation integration	Subsurface	Medium	8–12 weeks
G-09	No vendor marketplace / third-party app ecosystem	Platform	Low	16–20 weeks
G-10	No offline-first mobile sync for remote sites	Connectivity	Low	4–6 weeks
G-11	No automated production allocation (commingled wells)	Production	Low	4–6 weeks

1.2 Critical Gap Deep-Dives

G-01: Certified Historian. The platform currently uses InfluxDB for time-series storage and DuckDB for analytical queries. While technically capable, neither carries the industrial certification, vendor support ecosystem, or connector library of AVEVA PI System or Honeywell Uniformance PHD. Enterprise operators typically require a certified historian as

the system of record for regulatory compliance and insurance purposes. The recommended path is to implement a PI System AF (Asset Framework) compatible REST API layer over InfluxDB, allowing existing PI client tools (PI Vision, PI DataLink) to connect without replacing the underlying storage.

G-03: IEC 62443 Cybersecurity. The platform's security model (JWT/Keycloak, TLS, row-level security) is architecturally sound but has not been assessed against IEC 62443 Security Level 2 requirements. This standard is increasingly mandatory for offshore and critical infrastructure deployments. The gap is primarily documentation and process (security risk assessment, zone-and-conduit model, patch management procedures) rather than technical architecture changes.

G-04: Regulatory Reporting. U.S. operators must file API 14C (surface safety system documentation), BSEE OGOR (oil and gas operations report), and EPA Subpart W (greenhouse gas) reports. None of these are currently automated. The data exists in the platform's PostgreSQL schema, but the reporting templates, submission APIs, and audit trails are absent.

Part II: Improvement Recommendations

2.1 Priority 1 — Production Readiness (0–3 months)

The following improvements should be implemented before any production deployment. They address correctness, reliability, and operator trust rather than new features.

Real-time alarm acknowledgment persistence. Currently, alarm acknowledgments are in-memory state in the React UI. When the page refreshes, acknowledged alarms revert to active. The fix requires a PostgreSQL `alarm_acknowledgments` table (already in the schema) wired to the Go Alarm Manager service with a WebSocket broadcast on state change. This is a 2–3 day fix with significant operator trust impact.

Telemetry gap detection and backfill. The Rust edge agent buffers data during connectivity loss but the stream processor does not currently detect or flag telemetry gaps in the time-series. Operators need to distinguish “sensor reading 0” from “no reading received.” This requires a gap-detection job in the Python analytics service that marks InfluxDB time ranges as `DATA_QUALITY: GAP` and surfaces them in the Well Detail sensor charts as grey bands.

PostgreSQL connection pooling with PgBouncer. The current Go services use `pgx` connection pools, but under load (50+ concurrent users), direct connections to PostgreSQL will exhaust the `max_connections` limit. PgBouncer in transaction-pooling mode should be added to the Docker Compose stack as a sidecar, reducing connection overhead by 80–90%.

Structured logging and distributed tracing. The Go services use `log/slog` but do not emit OpenTelemetry traces. Without distributed tracing, debugging latency issues across the API Gateway → Well Management → Telemetry chain is extremely difficult. Adding `go.opentelemetry.io/otel` with a Jaeger or Tempo exporter is a 1-week effort with permanent operational benefit.

2.2 Priority 2 — Operator Experience (3–6 months)

Mobile field operations app. Field technicians need a native mobile experience for acknowledging alarms, logging observations, completing calibration work orders, and viewing well status without a laptop. A React Native app sharing the same API client layer as the web dashboard would cover 80% of field use cases. Offline-first sync using SQLite + background sync against the Go API is essential for remote sites with intermittent connectivity.

Shift handover report generation. At the end of each 12-hour shift, operators need a structured handover report summarizing: wells with active alarms, production vs. target, workovers in progress, and actions taken. This should be auto-generated by the Python analytics service at configurable intervals and delivered via email/Teams webhook. The data is already available; only the report template and delivery mechanism are missing.

Customizable dashboard layouts. The current Overview page has a fixed layout. Operators at different roles (production engineer, field supervisor, financial controller) need different KPI arrangements. A drag-and-drop dashboard builder using `react-grid-layout` would allow each user to save their preferred widget arrangement in PostgreSQL.

Dark/light theme toggle. The platform is currently dark-only. Outdoor tablet use in bright sunlight requires a high-contrast light theme. The CSS variable architecture already supports this; only the ThemeProvider toggle and light-mode color variables need to be added.

2.3 Priority 3 — Advanced Capabilities (6–12 months)

Physics-based digital twin integration. The ML pipeline currently uses purely data-driven models (XGBoost, LSTM). For well performance prediction, physics-based models (Nodal Analysis, IPR/VLP curves) provide better extrapolation outside the training data distribution. Integrating an open-source reservoir simulator (e.g., OPM Flow via Python subprocess) as a “digital twin” service would allow the platform to answer “what-if” questions: what happens to production if we increase ESP frequency by 5 Hz?

Automated production allocation. For commingled wells sharing a separator, the platform needs a production allocation module that distributes measured separator output back to individual wells using test separator data and allocation factors. This is a pure Python analytics service addition using the existing PostgreSQL schema.

Vendor marketplace. The platform’s architecture (open REST APIs, Kafka topics, PostgreSQL schemas) is well-suited to a third-party app ecosystem. A lightweight app registry allowing WT Petrotech customers to install certified third-party modules (e.g., a specialized corrosion monitoring module, a gas lift optimization module) would significantly expand the platform’s addressable market.

Part III: Competitive Comparison

3.1 Market Landscape

The oil and gas operations technology market is served by three distinct tiers of vendors: the **OT incumbents** (Emerson, Honeywell, Yokogawa, ABB) who sell DCS/SCADA systems with proprietary historian and HMI layers; the **industrial software platforms** (AVEVA, AspenTech, Seeq, Cognite) who sell analytics and visualization layers on top of existing historians; and the **oilfield services digital platforms** (SLB Delfi, Halliburton iEnergy, Weatherford ForeSite) who bundle digital tools with field services contracts. The OG-RMM Platform competes primarily in the second and third tiers.

3.2 Feature Comparison Matrix

The following table scores each platform across 20 capability dimensions on a 1–5 scale (5 = best-in-class, 1 = absent/minimal). Scores are based on publicly available product documentation, G2/Gartner Peer Insights reviews, and vendor datasheets as of Q1 2026.

Capability	OG-RMM	AVEVA PI System	Ignition SCADA	SLB Delfi	Cognite Data Fusion	Weatherford ForeSite	AspenTech PIMS
Real-time telemetry ingestion	5	5	4	4	4	4	2
OPC-UA / DNP3 / Modbus connectivity	4	5	5	3	3	4	2
MQTT / IoT edge agent	5	3	4	4	4	3	1
Time-series historian depth	3	5	4	4	4	3	3
ML / predictive maintenance	5	3	2	4	5	4	3
ESP failure prediction	5	2	1	4	3	5	1
ISA-18.2 alarm management	4	4	4	3	3	3	2
FPSO / subsea asset management	4	3	2	5	4	3	2
Financial ledger / cost tracking	5	1	1	2	2	2	3
Workover / intervention management	4	1	1	3	2	3	2
Calibration management	4	2	2	2	2	3	2
Geospatial / field map	4	2	3	4	5	3	1
Solar / low-bandwidth site support	4	2	3	3	2	3	1

Capability	OG-RMM	AVEVA PI System	Ignition SCADA	SLB Delfi	Cognite Data Fusion	Weatherford ForeSite	AspenTech PIMS
Multi-protocol edge (OPC-UA+DNP3+Modbus)	4	4	5	3	2	3	1
Regulatory reporting automation	1	3	2	3	2	3	4
Mobile field operations	1	3	3	4	3	4	2
Digital twin / physics simulation	2	2	1	5	4	3	5
IEC 62443 cybersecurity certification	1	4	3	4	3	3	3
Open API / extensibility	5	3	5	3	5	2	2
Deployment flexibility (cloud/on-prem/edge)	5	3	5	3	3	3	2
Total (max 100)	75	65	60	72	69	64	46

3.3 Platform-by-Platform Analysis

AVEVA PI System / Unified Operations Center. AVEVA PI System is the undisputed market leader for industrial time-series data management, with an installed base of over 20,000 sites globally. Its Asset Framework (AF) provides a hierarchical asset model that is deeply integrated with the broader AVEVA ecosystem (AVEVA Historian, AVEVA System Platform, AVEVA Unified Operations Center). The OG-RMM Platform's primary advantage over AVEVA is its **integrated financial ledger** (TigerBeetle double-entry accounting is absent from AVEVA), its **ML pipeline** (AVEVA's analytics rely on third-party tools like Seeq), and its **deployment cost** (AVEVA PI System licensing starts at approximately 50,000–200,000 per site, versus the OG-RMM Platform's open-source infrastructure stack). AVEVA's

primary advantage is its **historian certification**, **connector ecosystem** (3,000+ data source connectors), and **regulatory compliance** tooling.

Inductive Automation Ignition. Ignition is the most developer-friendly SCADA platform on the market, with unlimited tag licensing starting at approximately \$10,000 and a Python-scriptable environment. Its OPC-UA and Modbus connectivity is best-in-class. However, Ignition is fundamentally a **visualization and HMI platform**, not an analytics or ML platform. It has no built-in predictive maintenance, no financial ledger, no workover management, and no calibration scheduling. The OG-RMM Platform significantly outperforms Ignition in every analytics and business-process dimension. The two platforms are more complementary than competitive: Ignition could serve as the OPC-UA HMI layer feeding data into the OG-RMM Platform's Rust edge agent.

SLB Delfi. SLB's Delfi platform is the most capable competitor in the oilfield services tier, with deep subsurface modeling, well construction optimization, and production surveillance capabilities backed by SLB's 100+ years of domain expertise. Delfi's primary strengths are its **physics-based digital twin** (powered by Eclipse reservoir simulator), its **subsea asset management** (leveraging SLB's OneSubsea division), and its **regulatory compliance** tooling. The OG-RMM Platform matches Delfi on real-time telemetry, edge connectivity, and financial integration, and exceeds it on **open extensibility** and **deployment flexibility**. Delfi is a proprietary, cloud-only platform with pricing typically bundled into SLB service contracts, making it inaccessible to independent operators and smaller E&P companies — the primary market for the OG-RMM Platform.

Cognite Data Fusion. Cognite is the most technically sophisticated pure-software competitor, with an open industrial digital twin architecture, excellent geospatial capabilities, and strong ML/AI tooling. Cognite's primary differentiator is its **data contextualization engine** — it can automatically ingest and link data from multiple source systems (SAP, Maximo, PI System, SCADA) without manual mapping. The OG-RMM Platform's advantage over Cognite is its **end-to-end financial integration** (Cognite has no ledger or cost-tracking capability), its **workover and calibration workflow management**, and its **lower total cost of ownership** for greenfield deployments. Cognite is priced at approximately 200,000–500,000 per year for enterprise deployments.

Weatherford ForeSite. ForeSite is the most direct competitor to the OG-RMM Platform in the production optimization and ESP management space. ForeSite's ESP failure prediction models are trained on Weatherford's global ESP fleet data (millions of run-hours), giving it a significant **training data advantage** over the OG-RMM Platform's models. ForeSite also has a stronger **mobile field operations** capability and better **regulatory reporting** automation. The OG-RMM Platform's advantages are its **financial ledger integration**,

subsea/FPSO coverage, open architecture, and multi-protocol edge agent (ForeSite uses proprietary Weatherford edge hardware).

AspenTech PIMS. AspenTech’s Planning and Scheduling suite is primarily a **refinery and petrochemical** platform, not a wellhead/production platform. Its inclusion in this comparison is for completeness; it is not a direct competitor for upstream wellhead monitoring and control. AspenTech’s strength in LP optimization and production planning is unmatched, but it has minimal relevance to the WT Petrotech use case.

3.4 Pricing Comparison

Platform	Typical Entry Cost	Annual Licensing	Deployment Model
OG-RMM Platform	Open source infra + dev cost	\$0 licensing (self-hosted)	Cloud / on-prem / hybrid
AVEVA PI System	50,000–200,000 per site	15,000–60,000/site/yr	On-prem / AVEVA Cloud
Ignition SCADA	10,000–50,000 per site	3,000–15,000/site/yr	On-prem / cloud
SLB Delfi	Bundled with SLB services	500,000–2M+/yr enterprise	Cloud only
Cognite Data Fusion	\$200,000+	200,000–500,000/yr	Cloud only
Weatherford ForeSite	Bundled with Weatherford ESP	100,000–300,000/yr	Cloud / on-prem
AspenTech PIMS	\$500,000+	150,000–400,000/yr	On-prem / cloud

The OG-RMM Platform’s **zero licensing cost** is a structural competitive advantage for operators who have the engineering capability to deploy and operate it. The total cost of ownership over 5 years for a 50-well deployment is estimated at 200,000–500,000 (primarily engineering and cloud infrastructure), versus 1.5M–5M for comparable commercial platforms.

3.5 Unique Differentiators of the OG-RMM Platform

The following capabilities are either absent from or significantly weaker in all commercial competitors:

Double-entry financial ledger with TigerBeetle. No commercial O&G operations platform integrates a production-grade financial ledger. The OG-RMM Platform’s TigerBeetle integration enables real-time OPEX tracking, royalty settlement via Mojaloop, and production-to-revenue reconciliation within a single system. This eliminates the typical 2–4 week lag between production events and financial reporting.

Polyglot microservices with Rust edge agent. The Rust-based edge agent and stream processor provide memory-safe, sub-millisecond latency processing that is architecturally impossible in Python or Java-based competitors. For high-frequency vibration analysis (ESP bearing fault detection at 10 kHz sampling), this is a meaningful technical differentiator.

Open, self-hostable architecture. Every component of the OG-RMM Platform is open-source infrastructure (PostgreSQL, Redis, InfluxDB, Redpanda, Temporal, MinIO). Operators are not locked into a vendor’s cloud or pricing model. This is particularly valuable for national oil companies and operators in jurisdictions with data sovereignty requirements.

Part IV: Strategic Recommendations

4.1 Near-term (0–6 months): Close the Critical Gaps

The three gaps with the highest risk-adjusted priority are G-01 (historian certification), G-03 (IEC 62443), and G-04 (regulatory reporting). These are not feature additions but **trust and compliance prerequisites** for enterprise sales. A PI System compatibility layer (G-01) would allow the OG-RMM Platform to be positioned as a “PI System replacement” rather than a “PI System alternative,” dramatically expanding the addressable market.

4.2 Medium-term (6–12 months): Build the Moat

The platform’s unique combination of **financial ledger + ML pipeline + open architecture** is difficult for incumbents to replicate because it requires expertise across domains (distributed systems, ML, financial accounting, OT protocols) that no single commercial vendor currently combines. Deepening each of these three pillars — richer TigerBeetle financial analytics, more sophisticated ML models trained on real field data, and a richer protocol adapter library — will make the platform increasingly difficult to displace.

4.3 Long-term (12–24 months): Platform Ecosystem

The OG-RMM Platform’s open API architecture positions it well to become an **operator system of record** around which a third-party app ecosystem develops. The precedent is Ignition’s module marketplace, which has grown to 400+ third-party modules and is a significant source of competitive moat. Building a similar ecosystem — starting with WT Petrotech’s own specialized modules (pneumatic control, solar power management, coil tubing pressure pilots) — would create network effects that compound over time.

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